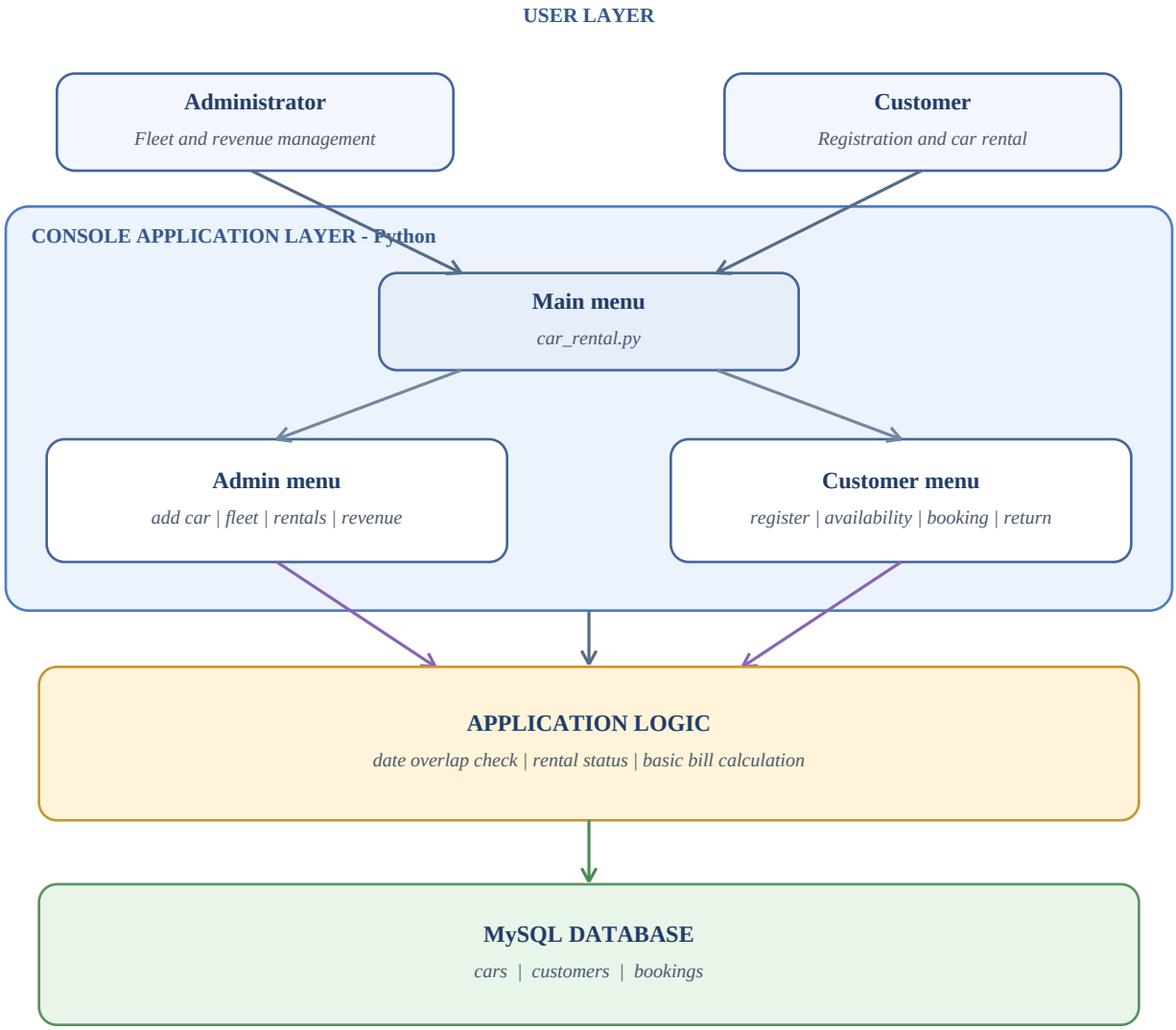
1. INTRODUCTION

The Car Rental System Using Python is a console-based application that helps a small rental shop manage its cars and customers. An administrator can add cars, view the fleet, see who has rented a car, and check collected revenue. Customers can register, check which cars are available for a date range, make or cancel a booking, and return a car. MySQL stores the car, customer, and booking records.

2. System Architecture Diagram



Data is saved in MySQL and read again when the menus need it.

3. Objectives

1. To manage the car fleet and its daily rental prices.
2. To let customers register and sign in with a simple account.
3. To check car availability for selected start and end dates.
4. To prevent two bookings for the same car from overlapping.
5. To record rentals, calculate a basic bill, and show revenue to the administrator.

4. Implementation Output Screenshots

Main menu (illustrative console output)

Car Rental System - Main Menu

```
========== CAR RENTAL SYSTEM ==========
1. Admin login  2. Customer register  3. Customer login  0. Exit
Choose: 1
Admin username: admin
Admin password: admin123
```

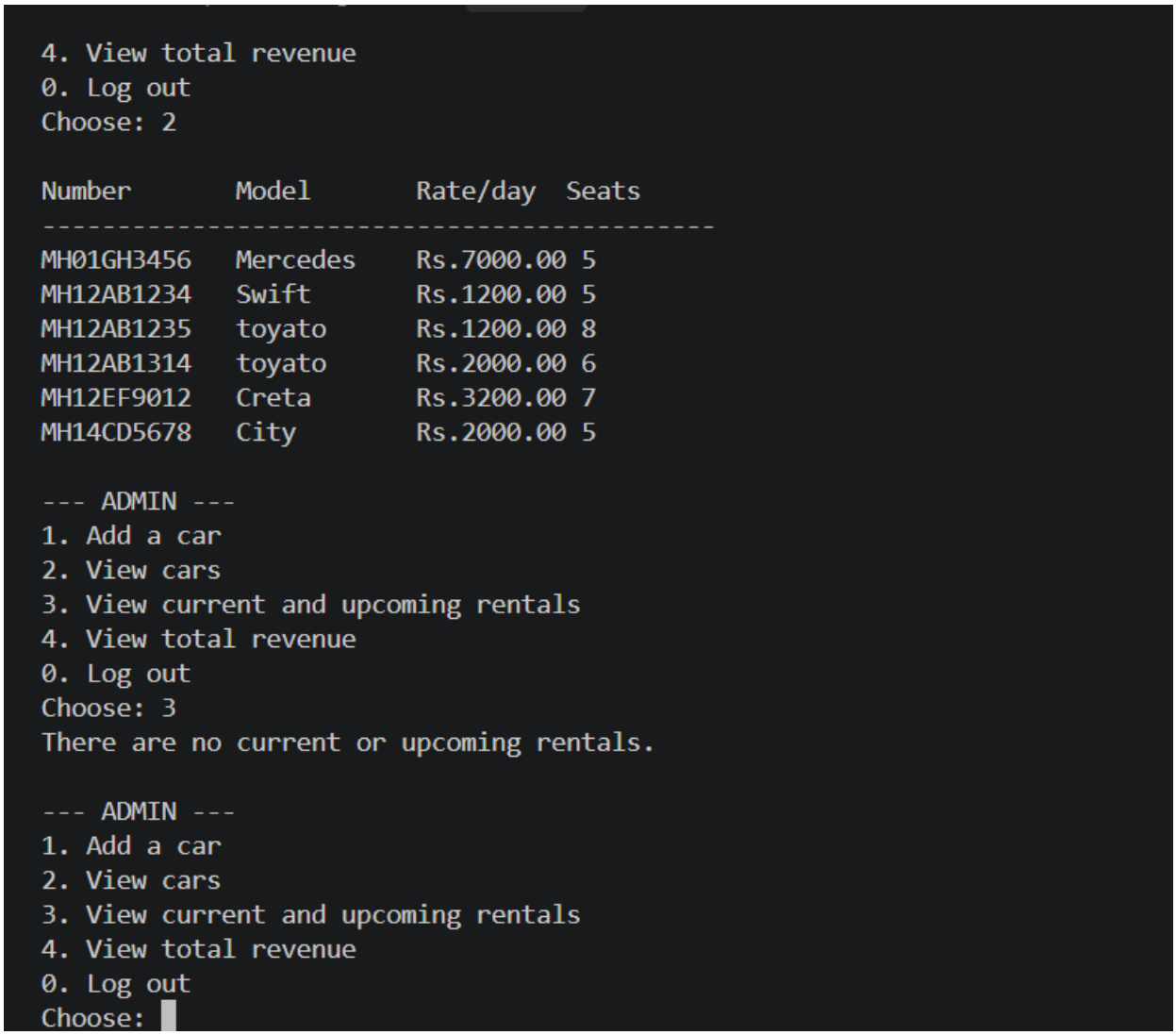
Customer registration (illustrative console output)

Customer Registration

```
--- REGISTER ---
Name: Asha Sharma
Phone: 9876543210
Choose username: asha
Choose password: ********
Account created. Choose customer login from the main menu.
```

4. Implementation Output Screenshots (Continued)

Uploaded terminal photo: fleet list and rental status



The screenshot shows the administrator menu, saved car list, and the message: 'There are no current or upcoming rentals.'

4. Implementation Output Screenshots (Continued)

Uploaded terminal photo: main menu, admin login, and fleet view

```
========== CAR RENTAL SYSTEM ==========
1. Admin login  2. Customer register  3. Customer login  0. Exit
Choose: 1
Admin username: admin
Admin password: admin123

--- ADMIN ---
1. Add a car
2. View cars
3. View current and upcoming rentals
4. View total revenue
0. Log out
Choose: 2

Number       Model       Rate/day  Seats
-------------------------------------------
MH01GH3456   Mercedes    Rs.7000.00 5
MH12AB1234   Swift       Rs.1200.00 5
MH12AB1235   toyato      Rs.1200.00 8
MH12AB1314   toyato      Rs.2000.00 6
MH12EF9012   Creta       Rs.3200.00 7
MH14CD5678   City        Rs.2000.00 5

--- ADMIN ---
1. Add a car
2. View cars
3. View current and upcoming rentals
```

Terminal names and menu labels shown in the uploaded photos:

Terminal Output

========== **CAR RENTAL SYSTEM** ==========

1. Admin login 2. Customer register 3. Customer login 0. Exit

Choose: 1

Admin username: admin

Admin password: admin123

--- ADMIN ---

1. Add a car

2. View cars

3. View current and upcoming rentals

4. View total revenue

0. Log out

5. Conclusion

The Car Rental System demonstrates how Python and MySQL can be used to organize the basic work of a rental shop. It provides separate administrator and customer menus, stores records in database tables, checks date ranges before confirming bookings, and calculates a simple rental bill when a car is returned. The administrator can also see current or upcoming rentals, the customer holding each car, and the revenue recorded from completed rentals. This project is a beginner-friendly example of using Python functions and SQL queries to solve a practical problem.